The mechanization of mental labour

The industrial revolution of the 18th and 19th century was about the mechanization of physical labour. In the many years of human history before that, there had been no violent change in the capabilities of humans, maybe except the discovery of fire. Yes, there was progress – but it was standard, simple progress. The industrial revolution was different however – it removed the need for man to physically toil in 12-hour shifts. Suddenly, labour was being economised faster than new uses of labour were being found and this caused great disruption in employment. Humanity, importantly, overcame this. Even as physical labour was mechanized, human agency became ever more important. Automated machinery required human agency – a human handler, who reported to a human manager and so on. Because we had more time on our hands, we found new economic activities like travel and entertainment to participate in. Goods were more economical and so was consumption, and humans moved on. Human agency was above the toil.

The internet has been touted as the other big change since then, but that – I argue – isn't true. Industrialization lifted human effort out of the physical and similarly, the internet lifted human logistics out of the physical. The internet allowed the non-physical exchange of information – in essence simply sharpening the tools of human agency. Artificial intelligence (AI) in the meantime, is different. AI, as is being discussed, is the mechanization of mental labour. The technological advancements of this century have enabled the replication of humanity's biggest differentiating advantage on the planet – its agency – in computers. We taught these systems to predict and write like a human would. Then, they we taught them how to *think* and *reason.* They now have their own *agency* – just like the average knowledge worker does. Given a problem statement, four of these 'agentic' systems can understand, strategize and develop a solution that they can also test, validate, execute and evaluate.[1]

The point I'm motivating here is that in the endeavour of economising the costs of our affluence, we first removed the requirement of physical effort by humans, and are now suggesting the removal of mental effort. Some, like Marc Andreessen, argue that this is the natural next step. That humans should leave all the physical and mental labour to technology systems and simply revel (by doing nothing) in the abundance such a reality would create.[2] Others shout against the forceful inclusion and regularization of AI by moneyed interests.[3] Either way, it would be sin to ignore the reality of today. It is humanity's collective duty then, to deliberately adopt individual and systemic considerations that cushion this violent adaption of our evolutionary capabilities that it will take to cope with.

Experiments to cope

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[1] Mucci, T. (2024, October 21). History of artificial intelligence. Ibm.com. https://www.ibm.com/think/topics/history-of-artificial-intelligence?utm_medium=OSocial&utm_source=Youtube&utm_content=WAIWW&utm_id=YT-201-History-of-AI

[2] Andreessen, M. (2023, October 16). The Techno-Optimist Manifesto. Andreessen Horowitz. https://a16z.com/the-techno-optimist-manifesto/

[3] Gioia, T. (2025, July 5). The Force-Feeding of AI on an Unwilling Public. Honest-Broker.com; The Honest Broker. https://www.honest-broker.com/p/the-force-feeding-of-ai-on-an-unwilling

Today, a knowledge worker interacts with their knowledge via a chat-box. To not read reports that run into hundreds of pages and instead rely on artificially generated "overviews" is becoming the norm in workplaces.[4] The instant generation of presentations and talking points, without tedious interaction with the knowledge (material) is increasingly acceptable. Logistically this obviously saves significant time. This knowledge worker doesn't need to 'sit with' the material anymore. They don't need to spend time reading it, understand the sequential arguments it makes, or draw inferences that may be relevant to parallel workstreams. Because of no direct interaction – and an artificial intelligence system's programmed tendency to be helpful,[5] the worker finds no points of friction or challenge in the material. In fact, to find friction and challenge the status quo (to *learn* something *new)*, humans historically have had to make an extra mental effort to overcome their inertia. This creates cognitive load; it takes time and sometimes even repeated attempts. In this manner, the brain learns, adapts, critiques – *thinks*.[6]

I posit that AI-enabled engagement with knowledge material is terrible for human critical thinking. Since we summarise, note key points and absorb *some* information from a lot of information, via an agreeable intermediary (like ChatGPT), we devoid our engagement with material of any friction – and thereby cognitive skills – and thinking critically. Status quo bias,[7] confirmation bias[8] and our general decision aversion results in our acceptance of what ChatGPT suggests about our material. It is hard to believe that an average Joe will dig deep into the material to fact-check, find gaps, or poke holes from that AI-powered excerpt. I hence suggest two distinct interventions that could abate this loss of critical thinking.

An i-frame intervention would address this reduction of critical thinking capabilities at an individual level via personalized information and training. I envision an experiment that allows us to measure differences in learning and critical thinking skills between two groups of participants – one that receives individual information and training, and one that doesn't. On average, should there be a comparable qualitative and quantitative difference in the groups' performance, we could ascertain the i-frame intervention to be relevant and applicable.

The setting is a graduate level class on, for example, Behavioural Economics. The total set of participants should be randomly assigned to two groups – AI Thinkers and Critical Thinkers. Both these groups are to engage with the course's knowledge material only via a screen-based learning portal that they are given special access to – that houses the material. This

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[4] TED - YouTube. (2025, December 28). How to Stop AI from Killing Your Critical Thinking | Advait Sarkar | TED. YouTube. https://www.youtube.com/watch?v=3lPnN8omdPA

[5] Cheng, M., Yu, S., Lee, C., Khadpe, P., Ibrahim, L., & Jurafsky, D. (2025). ELEPHANT: Measuring and understanding social sycophancy in LLMs. ArXiv (Cornell University). https://doi.org/10.48550/arxiv.2505.13995

[6] Understanding Brain Connectivity: How Our Minds Develop and Adapt. (2020). Brainbalancecenters.com. https://www.brainbalancecenters.com/blog/understanding-brain-connectivity-how-our-minds-develop-and-adapt

[7] Pilat, D., & Krastev, S. (2024). Status Quo Bias - Biases & Heuristics | The Decision Lab. The Decision Lab. https://thedecisionlab.com/biases/status-quo-bias

[8] Pilat, D., & Krastev, S. (2021). Why Do We Favor Our Existing Beliefs? The Decision Lab. https://thedecisionlab.com/biases/confirmation-bias

material cannot be interacted with outside of this medium, like in the case of Respondus's LockDown Browser. This implies that any and all concerning material should be available to students 24/7 via this portal – which cannot be downloaded and uploaded elsewhere. This is to prevent engagement with material using other intermediaries like paper prints or open-market AI chatbots. To establish a baseline measure of critical thinking skills, a general knowledge quiz is conducted for everyone before hand. The comparative measure of performance over the semester, as affected by the intervention is a graduate level quiz based on Bloom's Taxonomy's six levels[9] at the end of semester. The learning portal has the option to be have AI capabilities enabled. Without AI capabilities, the learning portal works simply as an environment where students can navigate between material documents and use key-word search-like functions.

As part of this i-frame intervention, both sets of participants receive straightforward and user-friendly information about AI-use's benefits and harms in learning from the most recent credible sources, in an unbiased way. An AI Thinkers' portal is AI-enabled as a default, and a Critical Thinkers' portal isn't AI-enabled as default. To uncheck the default setting, the participant must undertake tedious effort through the learning portal's settings menu and be reminded twice in that effort with information that re-enforces their categorization. This would mean that an AI Thinker switching out of the AI-enabled learning portal will receive information about the benefits of using AI on critical thinking and a Critical Thinker switching into the AI-enabled learning portal will receive information about the harms of using AI on critical thinking. A decision once made cannot be changed for the rest of the semester. The underlying idea here is that participants engage with material in a closed and controlled environment. This environment either mimics AI-chatbots of today's time, or the old-fashioned PDF viewer environment. We track the participant's decisions to keep/ uncheck their defaults and thereafter their performance in a cumulative quiz. This quiz maps the participant's performance across the six levels of Bloom's Taxonomy for 'understanding'. This mapping allows the experiment to measure individual and group averages (and their differences) of 'understanding' across six levels – which can be traced to the required application of critical thinking.[10] I posit there will be a negative difference in the understanding levels measured (and thereby critical thinking levels measured) for *remembering, analysing and creating* for AI Thinkers. *Evaluating* too may be more difficult for AI Thinkers, but not as much as these other three. Since the time and method of engagement per user will also be track-able, we could gain insight about methods of thinking more critically despite being grouped into AI Thinkers as well as less-helpful methods of engaging with material that worsen critical thinking from user behaviour in Critical Thinkers. This tracking of outliers could inform teaching methods of the future.

Another experiment that I believe could be useful uses a systemic intervention (or, s-frame) to possibly impact the effect AI has on critical thinking. This intervention uses a similar

[9] Bloom, B. (1956). BLOOM'S TAXONOMY. https://cft.vanderbilt.edu/wp-content/uploads/sites/59/Blooms-Taxonomy.pdf

[10] Westbrook, C. (2014, April 18). Teaching critical thinking using Bloom's Taxonomy. World of Better Learning | Cambridge University Press. https://www.cambridge.org/elt/blog/2014/04/18/teaching-critical-thinking-using-blooms-taxonomy/

setup. One behavioural economics graduate course, two groups split into AI Thinkers and Critical Thinkers. The same setup, except the defaults and individual-level notification about AI benefits/ harms is used here. Everyone undertakes a Bloom's Taxonomy based quiz that tests the participants general knowledge to establish a baseline. Everyone has access to the learning portal – from only where the knowledge material is accessible. In this scenario, both groups have to actively select between two options – AI-enabled and not-abled capabilities in the learning portal. As preliminary information in this case too, both sets of participants receive very straightforward and user-friendly information about AI-use's benefits and harms in learning from the most recent credible sources, in an unbiased way. The choice the participants make cannot be reversed for the rest of the semester. The catch for the Critical Thinkers is that they are again divided into two groups – Incentivized and Disincentivized. Both groups are given one additional 'extra-credit' question as part of their final assessment. The first group receives an incentive in the form of an additional extra-credit question at the end of their final assessment – to not choose the AI-enabled capability activation option in the learning portal. The second group is disincentivized against choosing the AI-enabled capabilities. Should they choose to use the AI-enabled version of the learning portal, they will be penalized with the single extra-credit question they have initially been allotted. The cumulative performance of the participant is measured in the same way as the first experiment – using a Bloom's Taxonomy based quiz to measure the understanding participant and map it to critical thinking abilities. The same tracking and testing setups from experiment are applied here. I posit we find that between the three groups, Disincentivized will show the greatest critical thinking abilities by outperforming Incentivized and AI Thinkers in Bloom's levels on Remembering, Analysing and Creating.

Is one approach better than another?

The i-frame intervention aims to nudge a Critical Thinker to accept the default or be treated with personalized information like 'harms of AI to critical thinking' and uses human tendencies to be loss averse to motivate the continued use of regular learning methods. This helps track i-frame intervention impacts on long-term critical thinking capabilities.

The s-frame intervention attempts to box the participant into choosing learning pathways that while seemingly tedious require and empower the use of critical thinking capacities – such that for the Critical Thinkers, a greater positive delta in critical thinking measures is observed. The intervention motivates the human tendency to be averse to penalties (losses) under systemic directives.

There do exist certain challenges in these experiments and they need to be refined given feedback. The usual issues of participant dropouts, inter-group leakages and exchanges of information may possibly undermine the findings. While I titled the groupings as AI Thinkers and Critical Thinkers for easier identification in experiment design, this titling may have confounding effects in motivations of the participants and affect their testing capabilities. To ensure that the experiment meets at least the basic degree of robustness, it is important that groupings are randomized. It is also important to conduct this in a course where *all* participants have actively elected to be in this course to ensure comparable baseline motivations. I also believe that the best test of critical thinking abilities may be suited for

subjects that have an equal split of quantitative and qualitative skills, like Physics, Psychology, Behavioural Economics etc. One constant consideration we should keep in mind is the loss of control about AI-use outside of the experimental setup given its accessibility and availability. This may hinder measurements of the critical thinking delta – and should be accounted for conservatively. It is important to note other efforts and develop the current body of work.[11]

The i-frame intervention uses the human status quo bias and the resulting decision aversion to adapt participant behaviour. It also reinforces information positioned to nudge the participant to remain in the original treatment setup motivating their aversion to loss (of critical thinking). The s-frame intervention is different – it has a directive approach, and 'penalizes' AI-use. This rules-based approach mimics a top-down policy that a school or institution may place on its students or an organization may place on its employees. Whether one is better than the other remains to be seen, but I imagine a mixed-method approach might work out best. To find this, the experiment should iteratively be tweaked with staged-interventions within the i- and s-frame approaches, but eventually also between them.

Artificial intelligence powered companions and chatbots are normal and have permeated substantial parts of the formal economy. Irrespective of the cost or benefit that brings to the world, it is important to measure, understand and address the uncertainty these developments create in the short run. It is imperative therefore to prepare human economies with awareness (in the form of research) and capabilities (in the form of tools) using the most efficient, but importantly also, effective interventions. As Keynes suggested almost a century ago,[12] we should mildly prepare and encourage experimenting in the arts of life and activities of purpose.

[11] Snow, J. (2025, June 26). AI Makes Research Easy. That Might Be a Problem. The Wall Street Journal. https://www.wsj.com/tech/ai/ai-learning-research-understanding-05fe0fde

[12] John Maynard Keynes, "Economic Possibilities for our Grandchildren (1930)," in Essays in Persuasion (New York: Harcourt Brace, 1932), 358-373

Draft

(the submission guideline suggested the submission of the final version as well as the draft version, and hence the following is pasted)

The mechanization of mental labour

The industrial revolution of the 18th and 19th century was about the mechanization of physical labour. In the many years of human history before that, there had been no violent change in the capabilities of humans, maybe except the discovery of fire. Yes, there was progress – but it was standard, simple progress. The industrial revolution was different however – it removed the need for man to physically toil in 12-hour shifts. Suddenly, labour was being economised faster than new uses of labour were being found and this caused great disruption in employment. In hindsight – what accumulated into the great depression in 1930 (1) – was just a blip. Humanity overcame this. Even as physical labour was mechanized, human agency became ever more important. Automated machinery required human agency – a human handler, who reported to a human manager and so on. Because we had more time on our hands, we found new economic activities like travel and entertainment to participate in. Goods were more economical and so was consumption, and humans moved on. Human agency was above the toil.

The internet has been touted as the other big change since then, but that - I argue - isn't true. Industrialization lifted human effort out of the physical and similarly, the internet lifted human logistics out of the physical. The internet allowed the non-physical exchange of information – in essence simply sharpening the tools of human agency. Artificial intelligence (AI) in the meantime, is different. AI, as is being discussed, is the mechanization of mental labour. The technological advancements of this century have enabled the replication of humanity's biggest differentiating advantage on the planet – its agency – in computers. We taught these systems to predict and write like a human would. Then, they we taught them how to *think* and *reason.* They now have their own *agency* – just like the average knowledge worker does. Given a problem statement, four of these 'agentic' systems can understand, strategize and develop a solution that they can also test, validate, execute and evaluate (2).

The point I'm motivating here is that in the endeavour of economising the costs of our affluence, we first removed the requirement of physical effort by humans, and are now suggesting the removal of mental effort. Some, like Marc Andreessen, argue that this is the natural next step. That humans should leave all the physical and mental labour to technology systems and simply revel (by doing nothing) in the abundance such a reality would create (3). Others shout against the forceful inclusion and regularization of artificial intelligence by moneyed interests (4). Either way, it would be sin to ignore the reality of today. It is humanity's collective duty then, to deliberately adopt individual and systemic considerations that cushion this violent adaption of our evolutionary capabilities that it will take to cope with.

Experiments to cope

Today, a knowledge worker interacts with their knowledge via a chat-box. To not read reports that run into hundreds of pages and instead rely on artificially generated

"overviews" is becoming the norm in workplaces (5). The instant generation of presentations and talking points, without tedious interacting with the knowledge (material) is increasingly acceptable. Logistically tracing such engagement, we see that this obviously saves significant time. This knowledge worker doesn't need to 'sit with' the material anymore. They don't need to spend time reading it, understand the sequential arguments it makes, or draw inferences that may be relevant to parallel workstreams. Because of no direct interaction – and an artificial intelligence system's programmed tendency to be helpful (6), the worker finds no points of friction or challenge in the material. In fact, to find friction and challenge the status quo (to *learn* something *new)*, humans historically have had to make an extra mental effort to overcome their inertia. This creates cognitive load; it takes time and sometimes even repeated attempts. In this manner, the brain learns, adapts, critiques – *thinks (7).*

I posit that such engagement with knowledge material is terrible for human critical thinking. Since we summarise, note key points and absorb *some* information from a lot of information, via an agreeable intermediary (like ChatGPT), we devoid our engagement with material of any friction – and thereby cognitive skills – to think critically. Status quo bias (8), confirmation bias (9) and our general decision aversion results in our acceptance of what ChatGPT suggests about our material. It is hard to believe that an average Joe will dig deep into the material to fact-check, find gaps, or poke holes from that given excerpt. I hence suggest two distinct interventions that could abate this loss of critical thinking.

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The setting is a graduate level class on, for example, Behavioural Economics. The total set of participants should be randomly assigned to two groups – AI Thinkers and Critical Thinkers. Both these groups are to engage with the course's knowledge material only via a screen-based learning portal that they are given special access to – that houses the material. This material cannot be interacted with outside of this medium. This implies that any and all concerning material should be available to students 24/7 via this portal – which cannot be downloaded and uploaded elsewhere. This is to prevent engagement with material using other intermediaries like paper prints or open-market AI chatbots. To establish a baseline measure of critical thinking skills, a general knowledge quiz is conducted for everyone. The comparative measure of performance over the semester and as affected by the intervention is a graduate level quiz based on Bloom's Taxonomy's six levels (10) at the end of semester. The learning portal has the option to be have AI capabilities enabled. Without AI capabilities, the learning portal works simply as an environment where students can navigate between material documents and use key-word search-like functions.

As part of the i-frame intervention. As preliminary information, both sets of participants receive very straightforward and user-friendly information about AI-use's benefits and harms

in learning from the most recent credible sources, in an unbiased way. An AI Thinkers' portal is AI-enabled as a default, and a Critical Thinkers' portal isn't AI-enabled as default. To uncheck the default setting, the participant must undertake tedious effort through the learning portal's settings menu and be reminded twice in that effort with information that re-enforces their categorization. This would mean that an AI Thinker switching out of the AI-enabled learning portal will receive information about the benefits of using AI on critical thinking and a Critical Thinker switching into the AI-enabled learning portal will receive information about the harms of using AI on critical thinking. A decision once made cannot be changed for the rest of the semester. The underlying idea here is that participants engage with material in a closed and controlled environment. This environment either mimics AI-chatbots of today's time, or the old-fashioned PDF viewer environment. We track the participant's decisions to keep/ uncheck their defaults and thereafter their performance in a cumulative quiz. This quiz maps the participant's performance across the six levels of Bloom's Taxonomy for 'understanding'. This mapping allows the experiment to measure individual and group averages (and their differences) of 'understanding' across six levels – which can be traced to the required application of critical thinking (11). I posit there will be a negative difference in the understanding levels measured (and thereby critical thinking levels measured) for *remembering, analysing and creating* for AI Thinkers*. Evaluating* too may be more difficult for AI Thinkers, but not as much as these other three. Since the time and method of engagement per user will also be track-able, we could gain insight about methods of thinking more critically despite being grouped into AI Thinkers as well as less-helpful methods of engaging with material that worsen critical thinking from user behaviour in Critical Thinkers. This tracking of outliers could inform teaching methods of the future.

Another experiment that I believe could be useful uses a systemic intervention (or, s-frame) to possibly impact the effect AI has on critical thinking. This intervention uses a similar setup. One behavioural economics graduate course, two groups split into AI Thinkers and Critical Thinkers. The same setup, except the defaults and individual-level notification about AI benefits/ harms is used here. Everyone undertakes a Bloom's Taxonomy based quiz that tests the participants general knowledge to establish a baseline. Everyone has access to the learning portal – from only where the knowledge material is accessible. In this scenario, both groups have to actively select between two options – AI-enabled and not-abled capabilities in the learning portal. As preliminary information in this case too, both sets of participants receive very straightforward and user-friendly information about AI-use's benefits and harms in learning from the most recent credible sources, in an unbiased way. The choice the participants make cannot be reversed for the rest of the semester. The catch for the Critical Thinkers is that they are again divided into two groups – Incentivized and Disincentivized. The first group receives an incentive in the form of a single-grade upgrade to their cumulative performance at the end – to not choose the AI-enabled capability activation option in the learning portal. The second group is disincentivized against choosing the AI-enabled capabilities. Should they choose to use the AI-enabled version of the learning portal, they will be penalized with a single-grade downgrade. The cumulative performance of the participant is measured in the same way as the first experiment – using a Bloom's Taxonomy based quiz to measure the understanding participant and map it to critical

thinking abilities. The same tracking and testing setups from experiment are applied here. I posit we find that between the three groups, Disincentivized will show the greatest critical thinking abilities by outperforming Incentivized and AI Thinkers in Bloom's levels on Remembering, Analysing and Creating.

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There do exist certain challenges in these experiments. Notwithstanding the potential contention of an institutional review board to allow unassuming students to be recruited as participants, these experiments need to be refined given feedback. The usual issues of participant dropouts, inter-group leakages and exchanges of information may possibly undermine the findings. While I titled the groupings as AI Thinkers and Critical Thinkers for easier identification in experiment design, this titling may have confounding effects in motivations of the participants and affect their testing capabilities. To ensure that the experiment meets at least the basic degree of robustness, it is important that groupings are randomized. It is also important to conduct this in a course where *all* participants have actively elected to be in this course to ensure comparable baseline motivations. I also believe that the best test of critical thinking abilities may be suited for subjects that have an equal split of quantitative and qualitative skills, like Physics, Psychology, Behavioural Economics etc. One constant consideration we should keep in mind is the loss of control about AI-use outside of the experimental setup given its accessibility and availability. This may hinder measurements of the critical thinking delta – and should be accounted for conservatively.

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